

Computer Science Tutorial — ENS Year 2 — Session 09 (EN)

****Instructor:**** BELACEL Madani

****Level:**** ENS — Year 2 — English Department

****Duration:**** 1h30

Exercise 1 — Types of networks

****Statement:**** Associate LAN, MAN, WAN with: computer room, city network, global Internet.

****Detailed correction:****

- Computer room → LAN (Local Area Network).
- City network → MAN (Metropolitan Area Network).
- Worldwide Internet → WAN (Wide Area Network).

Exercise 2 — Topologies

****Statement:**** Draw a star topology with 4 stations and a central switch. Name a benefit.

****Detailed correction:****

- 4 PCs individually connected to the switch (central hub).
- Advantage: failure of a cable only affects one station; easy troubleshooting.
- Very common in Wi-Fi (access point = center).

Exercise 3 — Network components

****Statement:**** Connect component and role: switch, router, network card, Ethernet cable.

****Detailed correction:****

- Switch: interconnects machines via LAN.
- Router: connects LAN to the Internet (IP routing).
- Network card (NIC): machine/network interface.
- Ethernet cable: wired transmission medium.

Exercise 4 — Addressing and DHCP

****Statement:**** A station displays `169.254.x.x`. What likely problem? Which protocol resolves IP assignment?

****Detailed correction:****

- APIPA address: no DHCP server reachable (Wi-Fi not connected or box turned off).
- DHCP must assign IP, mask, gateway, DNS.
- Solution: check Wi-Fi/Ethernet connection, restart box, contact IT support.

Exercise 5 — Campus case

****Statement:**** Propose 3 rules for good use of the university network.

****Detailed correction:****

1. Educational and research use (no illegal downloads).
2. Do not share institutional Wi-Fi credentials.
3. Respect the IT charter (security, bandwidth, shared printers).

Session progress — 1h30 (90 min)

Phase	Duration	Cumulative total
Welcome	5 mins	5 mins
Exercise 1 — Types of networks	15 mins	20 mins
Exercise 2 — Topologies	18 mins	38 mins
Exercise 3 — Components	18 mins	56 mins
Exercise 4 — DHCP	15 mins	71 mins
Exercise 5 — Campus charter	14 mins	85 mins
Correction	5 mins	90 mins
Total	**90 mins**	